

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2019 – 2023	B.S in Physics	Indian Institute of Technology, Kanpur	6.4/10
2023 – 2024	Second Major - B.Tech Aerospace	Indian Institute of Technology, Kanpur	6.4/10
2018	Class XII (ISC)	St. Joseph’s Convent School, Varanasi	87.5%

PROFESSIONAL EXPERIENCE

TerraNxt | Gurgaon, India | Founding Software Engineer (ScadaNxt) (Nov ’25 – Present)

- **Solo Full-Stack Development:** Architected and built an end-to-end SCADA and O&M platform from scratch as sole engineer, transitioning the company from reliance on third-party monitoring vendors (Cleanmax, Quadrical) to an indigenous solution.
- **Distributed Data Architecture:** Designed a containerized telemetry pipeline using **MQTT**, **Kafka**, and a multi-tier database (**InfluxDB** for time-series, **PostgreSQL** for relational data, **MinIO** for object storage) for multi-plant portfolio management.
- **Physics-Based Simulation Engine:** Built a validation layer using **PvLib** and **PyBaMM** that maintains a digital twin of each plant, continuously comparing simulated expected performance against real-time telemetry to detect anomalies and degradation.
- **Computer Vision Pipeline:** Developed a drone imagery processing system using **UNet** and **SegFormer** for solar panel detection (80% accuracy), automatically generating 3D spatial models for asset tracking and defect identification.
- **AI-Powered Analytics:** Integrated an intelligent assistant using **LangGraph** that translates natural language queries into SQL/Flux for real-time diagnostics, enabling voice and chat-based performance analysis.
- **Predictive Forecasting:** Implemented a generation forecasting engine correlating **NSRDB weather data** with plant-specific telemetry, providing stakeholders with predictive yield estimates and performance benchmarking against expected output.
- **Internal Validation:** Validated system accuracy against real plant data, successfully matching daily generation reports (PR, CUF, efficiency calculations) within industry-acceptable margins as confirmed by domain experts.

Smarttrak AI | Research Scientist & Lead Product Engineer (May ’24 – Nov ’25)

- **Pilot Deployment Leadership:** Led technical development and on-site demonstrations for the company’s first solar plant pilot project, working with a remote team of 5-6 engineers to validate core product capabilities with potential enterprise clients.
- **IoT Control System:** Engineered a production telemetry stack (**MQTT** → **Kafka** → **TimescaleDB**) for a 20-30 tracker installation, developing real-time control software that synchronized hardware movements with solar position algorithms.
- **Physics-Informed Neural Networks:** Designed experimental PINNs in **PyTorch** for 1250kW solar inverters, embedding power flow equations into loss functions to estimate useful life and reconstruct states from sparse SCADA readings.
- **Satellite-Scale Computer Vision:** Trained convolutional autoencoders on **A100 GPU clusters** to compress satellite irradiance maps, enabling geographic-scale solar forecasting with mixed-precision training acceleration.
- **3D Reconstruction Pipeline:** Leveraged **NVIDIA Instant-NGP** and NeRFs to generate volumetric terrain models from satellite imagery, creating interactive 3D visualizations in **Three.js** for client demonstrations and solar siting analysis.
- **Microgrid Energy Management:** Designed a modular EMS framework using **pvlb** and **PyBaMM** for hybrid (PV+Battery+DG) systems, implementing predictive orchestration logic for islanding scenarios and DG synchronization transitions.
- **Cloud Infrastructure:** Developed **FastAPI** microservices containerized with **Docker** on **Azure VMs**, implementing CI/CD pipelines that reduced deployment failures and improved system reliability through automated testing and rollback mechanisms.

ACHIEVEMENTS

Academic:

- Second Major in Aerospace Eng. (**B.Tech AE**) | IIT Kanpur | 2024
- Recipient of the **C3i Research scholarship**, awarded for research in cyber security | C3i labs, IIT Kanpur | 2021
- Secured an **All India Rank 4412** in **Joint Entrance Examination Advanced 2019** among 2.45 Lakh shortlisted candidates
- Secured an **All India percentile 98.9253** in **Joint Entrance Examination Main 2019** among 1.2 million candidates

PROJECTS

Mighty Resume | Solo-built SaaS · mightyresume.in | September 2025

- Built and deployed an AI-powered resume analysis SaaS that evaluates resumes across **9 ATS-focused dimensions** using GPT-4o-mini, generating structured scoring, section-wise feedback, grading, and recommendations through a custom Flask backend
- Designed and managed the complete **production-grade infrastructure** on Azure Kubernetes Service (AKS) — including NGINX Ingress, auto-renewing HTTPS/TLS via Let’s Encrypt, autoscaling, routing, secrets management, and high-availability deployment.
- Engineered a **fully automated zero-downtime CI/CD pipeline** using GitHub Actions, Docker, Azure Container Registry, and Kubernetes rolling deployments; integrated automated testing, multi-architecture builds, and production rollout orchestration
- Delivered the entire platform **single-handedly within 30 days** on limited cloud credits, with modular service-oriented architecture designed for future expansion into multilingual AI interview systems, premium subscriptions, and scalable marketplace features